Date: 24/10/2026

C02 - ASSESSMENT TOOL 2

SCENARIO BASED

Course Code: DSA03

Course Title: Natural Language Processing

Reg.no: 192425005 **Name:** S. Yuva Guru Mani Varma

---

1. Morphological Processing for Search Engine Indexing

Input words: analyzing, analysis, analytical

Python program

```
words = ["analyzing", "analysis", "analytical"]

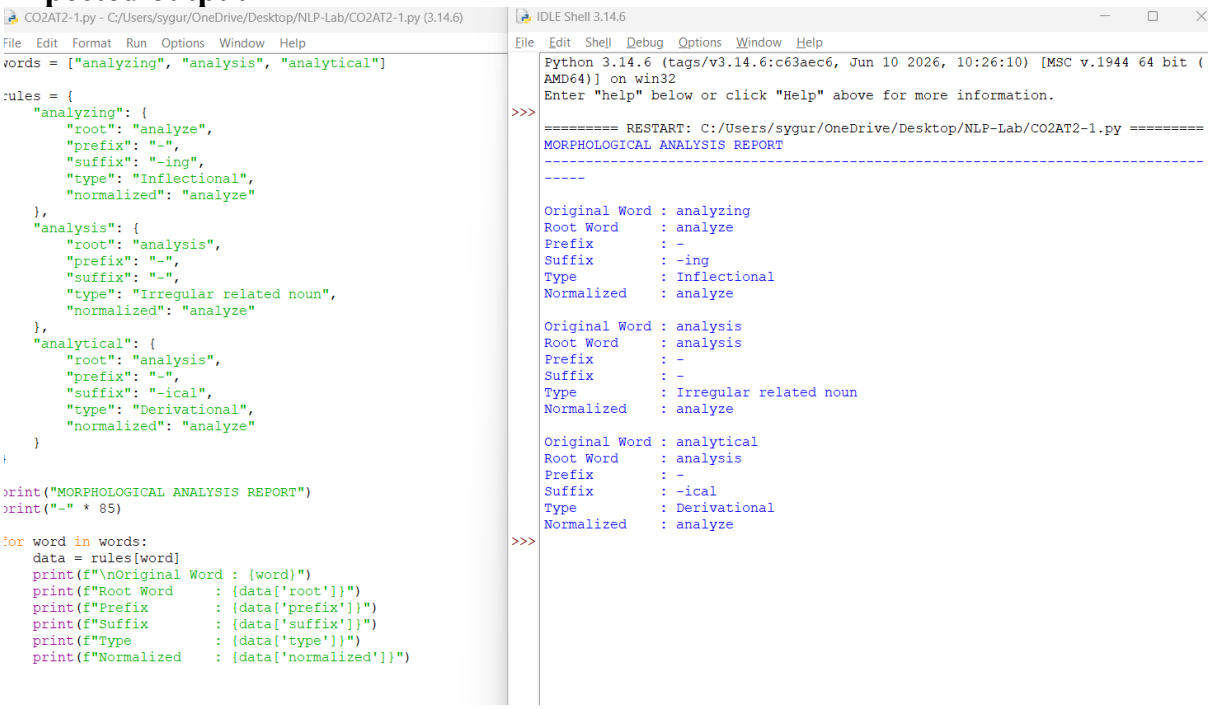
rules = {
   "analyzing": {
      "root": "analyse",
      "prefix": "-",
      "suffix": "-ing",
      "type": "Inflectional",
      "normalized": "analyse"
   },
   "analysis": {
      "root": "analysis",
      "prefix": "-",
      "suffix": "-",
      "type": "Irregular related noun",
      "normalized": "analyse"
   },
   "analytical": {
      "root": "analysis",
      "prefix": "-",
      "suffix": "-ical",
      "type": "Derivational",
      "normalized": "analyse"
   }
}

print("MORPHOLOGICAL ANALYSIS REPORT")
print("-" * 85)

for word in words:
   data = rules[word]
   print(f"\nOriginal Word : {word}")
   print(f"Root Word     : {data['root']}")
   print(f"Prefix        : {data['prefix']}")
```

```
print(f'Suffix       : {data['suffix']}")
print(f'Type         : {data['type']}")
print(f'Normalized   : {data['normalized']}")
```

Expected output



Result: All three terms are mapped to the common indexing representation analyse, allowing related search results to be grouped together.

Note: Analysis is an irregularly related noun, not a regular suffix-based inflection of analyse. This program uses a predefined lexical mapping to handle that relationship.

2. Morphological Parser for Sentiment Analysis

Input words: disagree, agreement, agreeable

Python program

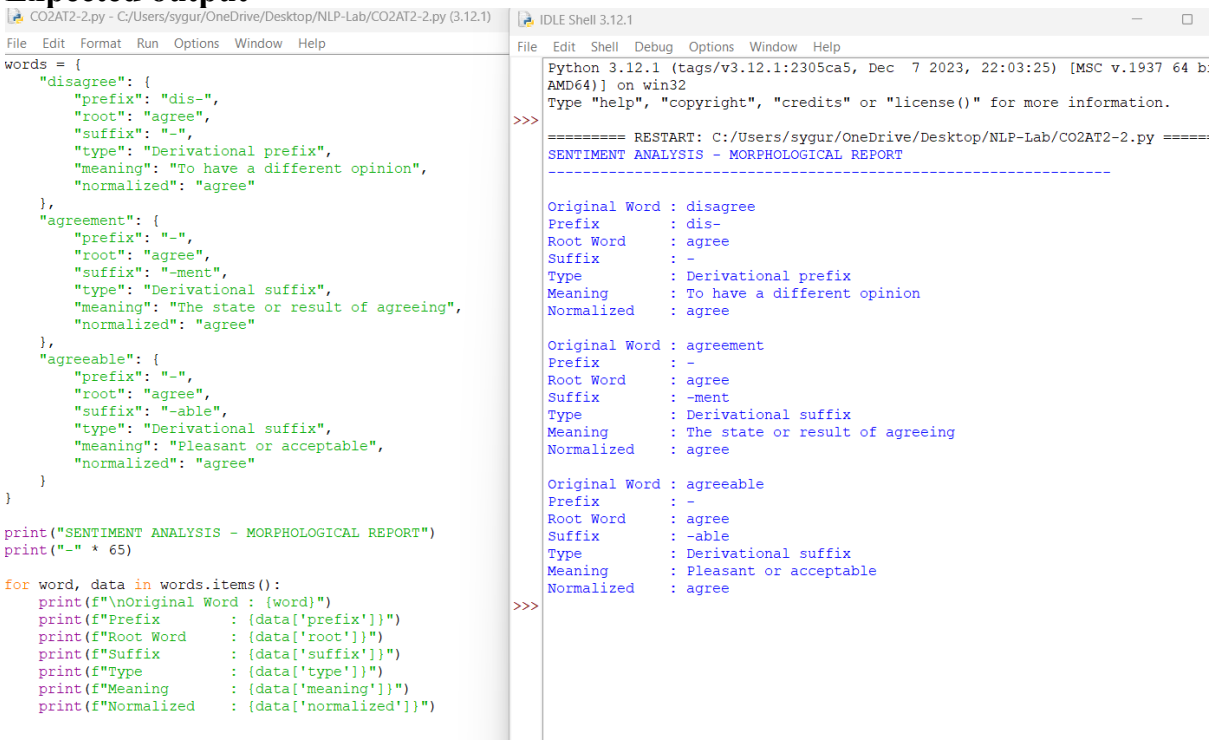
```
words = {
  "disagree": {
    "prefix": "dis-",
    "root": "agree",
    "suffix": "-",
    "type": "Derivational prefix",
    "meaning": "To have a different opinion",
    "normalized": "agree"
  },
  "agreement": {
    "prefix": "-",
    "root": "agree",
    "suffix": "-ment",
    "type": "Derivational suffix",
    "meaning": "The state or result of agreeing",
    "normalized": "agree"
```

```
    },
    "agreeable": {
        "prefix": "-",
        "root": "agree",
        "suffix": "-able",
        "type": "Derivational suffix",
        "meaning": "Pleasant or acceptable",
        "normalized": "agree"
    }
}

print("SENTIMENT ANALYSIS - MORPHOLOGICAL REPORT")
print("-" * 65)

for word, data in words.items():
    print(f"\nOriginal Word : {word}")
    print(f"Prefix        : {data['prefix']}")
    print(f"Root Word     : {data['root']}")
    print(f"Suffix        : {data['suffix']}")
    print(f"Type          : {data['type']}")
    print(f"Meaning       : {data['meaning']}")
    print(f"Normalized    : {data['normalized']}")
```

Expected output



Result: All words are normalized to agree. The prefix dis- reverses the meaning, while -ment and -able create new word forms. Sentiment analysis must consider these meaning changes rather than treating all three words as having identical sentiment.

3. Morphology-Based Normalization for News Aggregation

Input words: govern, government, governance

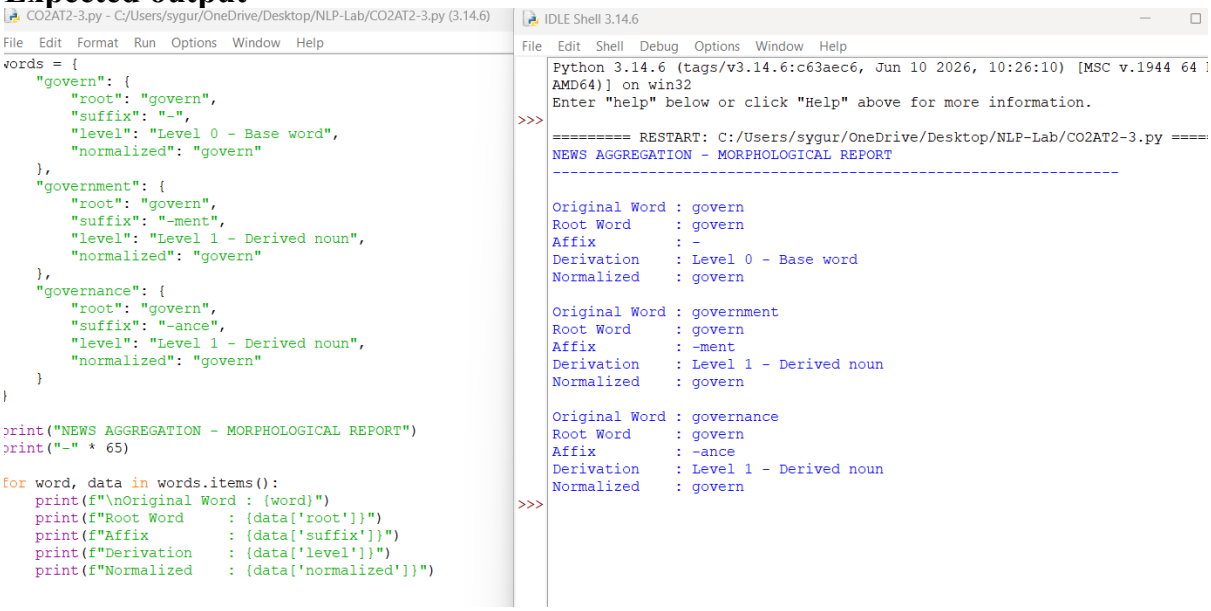
Python program

```
words = {
   "govern": {
      "root": "govern",
      "suffix": "-",
      "level": "Level 0 - Base word",
      "normalized": "govern"
   },
   "government": {
      "root": "govern",
      "suffix": "-ment",
      "level": "Level 1 - Derived noun",
      "normalized": "govern"
   },
   "governance": {
      "root": "govern",
      "suffix": "-ance",
      "level": "Level 1 - Derived noun",
      "normalized": "govern"
   }
}

print("NEWS AGGREGATION - MORPHOLOGICAL REPORT")
print("-" * 65)

for word, data in words.items():
   print(f"\nOriginal Word : {word}")
   print(f'Root Word     : {data['root']}")
   print(f'Affix         : {data['suffix']}")
   print(f'Derivation    : {data['level']}")
   print(f'Normalized    : {data['normalized']}")
```

Expected output



Result: The three words are mapped to govern. The suffixes -ment and -ance form nouns from the base verb. Their normalized forms can be used to group related news documents for topic modeling and clustering.

4. Morphological Parsing for Document Classification

Input words: activate, activation, reactivation

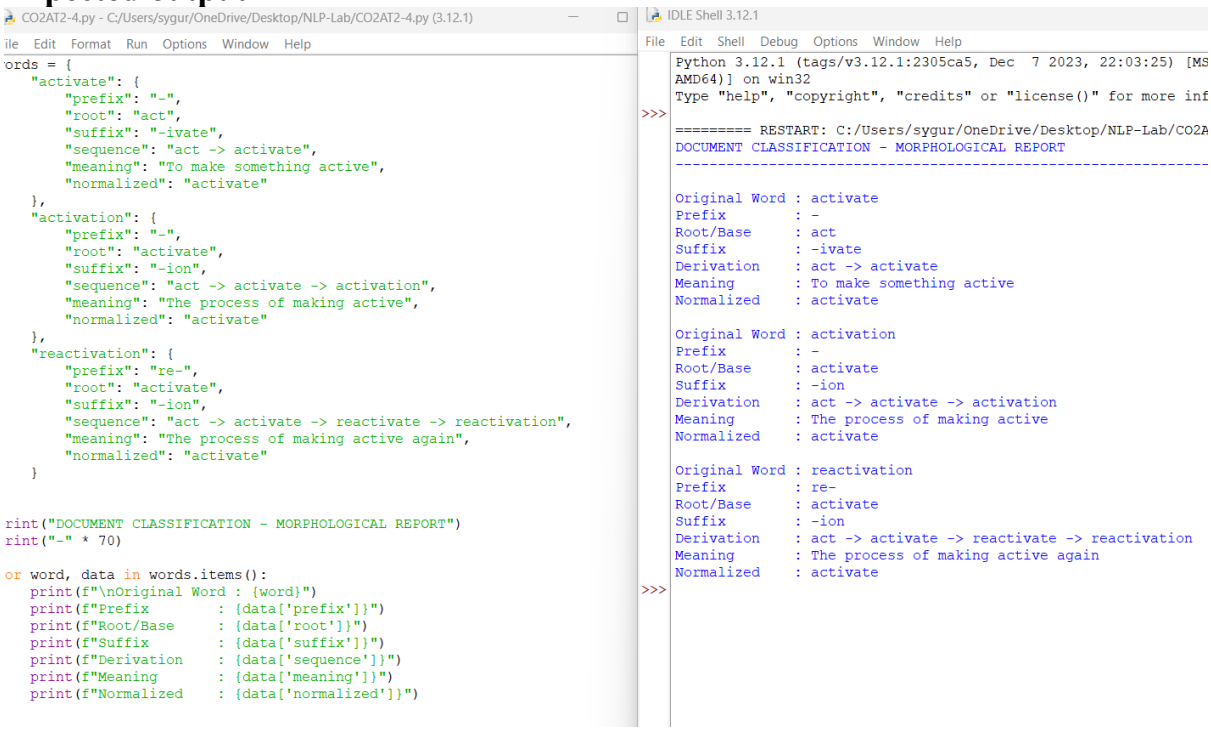
Python program

```
words = {
   "activate": {
      "prefix": "-",
      "root": "act",
      "suffix": "-ivate",
      "sequence": "act -> activate",
      "meaning": "To make something active",
      "normalized": "activate"
   },
   "activation": {
      "prefix": "-",
      "root": "activate",
      "suffix": "-ion",
      "sequence": "act -> activate -> activation",
      "meaning": "The process of making active",
      "normalized": "activate"
   },
   "reactivation": {
      "prefix": "re-",
      "root": "activate",
      "suffix": "-ion",
      "sequence": "act -> activate -> reactivate -> reactivation",
      "meaning": "The process of making active again",
      "normalized": "activate"
   }
}

print("DOCUMENT CLASSIFICATION - MORPHOLOGICAL REPORT")
print("-" * 70)

for word, data in words.items():
   print(f"\nOriginal Word : {word}")
   print(f"Prefix        : {data['prefix']}")
   print(f"Root/Base     : {data['root']}")
   print(f"Suffix        : {data['suffix']}")
   print(f"Derivation    : {data['sequence']}")
   print(f"Meaning       : {data['meaning']}")
   print(f"Normalized    : {data['normalized']}")
```

Expected output



Result: The words are normalized to activate. The prefix re- indicates repetition, while -ion converts the verb into a noun. This helps a document classification system group related concepts.
Note: act + -ivate is a simplified morphological representation; activate is best treated as the established base verb in practical morphological parsing.

5. Inflectional Morphology for Search Optimization

Input words: create, creates, creating

Python program

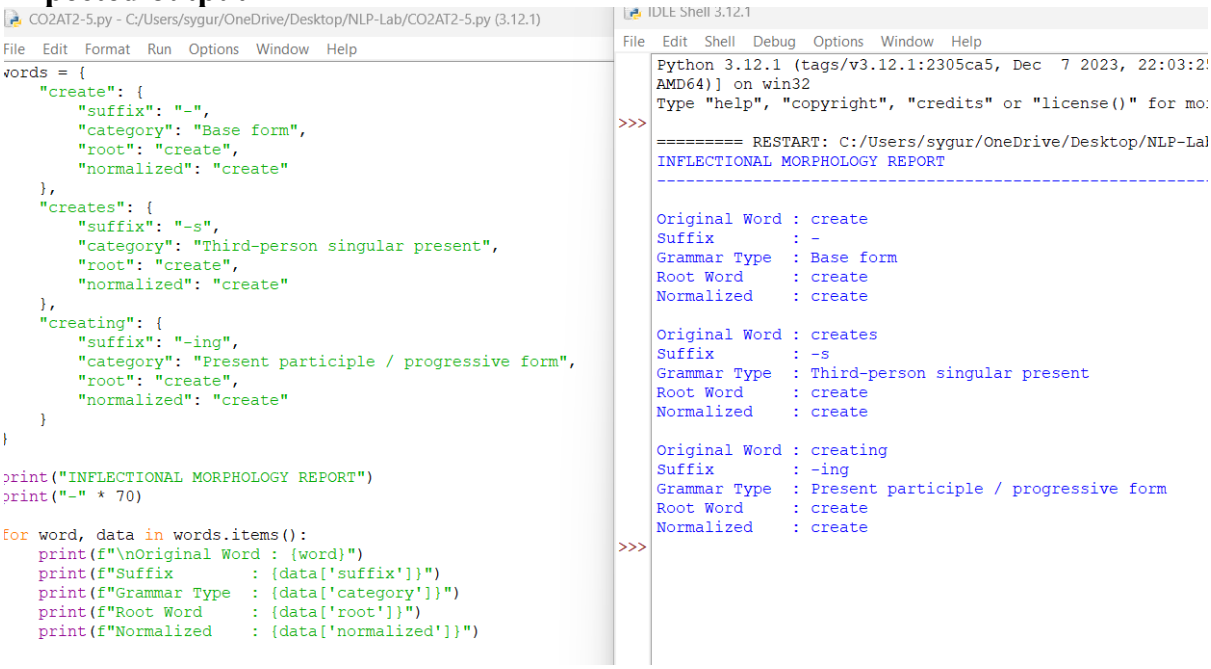
```
words = {
   "create": {
      "suffix": "-",
      "category": "Base form",
      "root": "create",
      "normalized": "create"
   },
   "creates": {
      "suffix": "-s",
      "category": "Third-person singular present",
      "root": "create",
      "normalized": "create"
   },
   "creating": {
      "suffix": "-ing",
      "category": "Present participle / progressive form",
```

```
        "root": "create",
        "normalized": "create"
    }
}

print("INFLECTIONAL MORPHOLOGY REPORT")
print("-" * 70)

for word, data in words.items():
    print(f"\nOriginal Word : {word}")
    print(f'Suffix        : {data['suffix']}")
    print(f'Grammar Type  : {data['category']}")
    print(f'Root Word     : {data['root']}")
    print(f'Normalized    : {data['normalized']}")
```

Expected output



Result: All three words are normalized to create. The suffix -s indicates third-person singular present tense, and -ing indicates a participial form used in progressive constructions. These are inflectional changes that preserve the core lexical meaning.